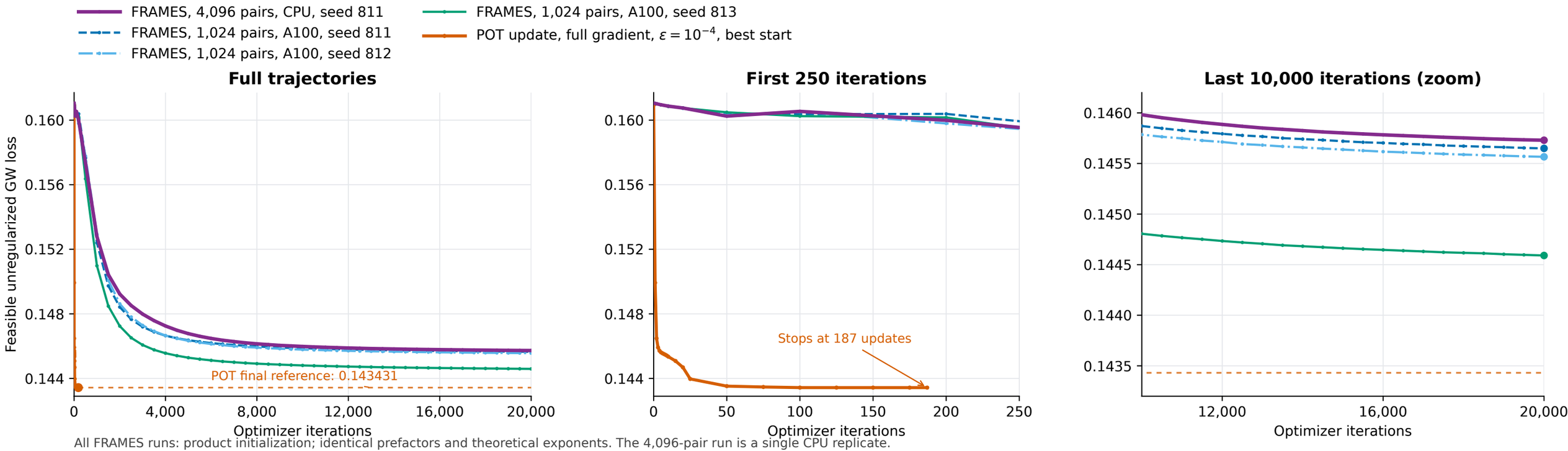
BMMC: feasible GW loss versus optimizer iterations

6,224 × 6,224 coupling • FP64 • Lower loss is better



All FRAMES runs: product initialization; identical prefactors and theoretical exponents. The 4,096-pair run is a single CPU replicate.

POT: best of 12 completed entropy/initialization trials; permutation-mixture start, seed 903. Each optimizer update has a different computational cost.

Dots are measured diagnostics; connecting lines are unsmoothed. The dashed POT line is its final-loss reference, not additional iterations.

CPU and CUDA random streams differ, including when the numerical seed is the same; this comparison does not isolate the effect of batch size.